

We measure it.



Highly efficient flue gas analyzer

testo 320 – Just a few “clicks” away from a heating system diagnosis

High-resolution colour graphic display

Quick and easy menu structure

Storage space for 500 measurement values

Measurement of flue gas, draught, pressure, ambient CO, differential temperature and gas leak detection

O₂ and CO sensor and flue gas probe with temperature probe

TÜV-tested according to EN 50379, Parts 1-3



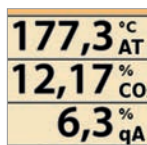
The new testo 320 is a high-quality measuring instrument for efficient flue gas analysis. Its wide measuring range makes it a reliable partner for eliminating malfunctions and emergencies, monitoring legal limit values or for daily routine work servicing heating systems. The numerous measurement menus of the testo 320 are clearly structured.

Standardized menu procedures, which are stored in the instrument specifically for your country, simplify operation – depending on which standards you are dealing with. The high-resolution display allows a detailed presentation of the measurement procedures and is easily legible even under the worst conditions.

Product properties

High-resolution colour graphic display

The measurement menus and measurement values are presented in detail and always easily legible.



Sensor monitoring

Integrated traffic light system which continuously monitors the sensor functionality.



Fast sensor zeroing

Automatic zeroing of the sensor in only 30 seconds after start-up, and which can be cancelled if not required.



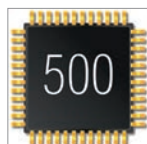
Sensors exchangeable by the user

Easy exchange of sensors by the user – no adjustment necessary.



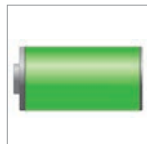
Memory

Up to 500 measurement protocols can be saved and called up in the memory of the testo 320.



Lithium battery

Operation with a Lithium battery (1500mAh) – no battery change necessary, up to eight hours running time, charging via USB connection possible.



Attachment

Integrated magnets for fast attachment to burner/boiler.



**Stamp of approval**

The flue gas analyzer testo 320 is TÜV-tested according to EN 50379, Parts 1-3.

**Robust design**

Robust, durable instrument – ideally suited even to rough surroundings.

**Condensate trap**

Integrated condensate trap – very easily emptied.

**Efficient exchange of probes**

Fast and easy exchange of probes via the probe coupling. All gas paths are connected to the instrument at once with the bayonet connection.

**Probe filter**

Easy exchange of probe filter.

**Flexibility with modular probes**

A range of probe lengths and diameters ensure a high degree of flexibility for all applications. To exchange the probe shaft, it is simply placed on the probe handle and engages.

Ordering data / Accessories

testo 320 set for chimneysweeps

0632 3220	testo 320 with H ₂ -compensated CO sensor and Bluetooth
0554 1105	USB mains unit
0600 9761	Flue gas probe modular (length 300 mm, Ø 8 mm)
0600 9787	Combustion air probe (length 190 mm)



testo 320 set for heating constructors

0632 3220	testo 320 with H ₂ -compensated CO sensor
0554 1105	USB mains unit
0516 3300	System case (height: 130 mm)
0554 0549	Testo fast printer IRDA
0600 9741	Compact flue gas probe (length 300 mm, Ø 6 mm)
0600 9787	Combustion air probe (length 190 mm)



Measuring instrument with options	Part no.
testo 320 flue gas analyzer, incl. O ₂ sensor, calibration protocol, graphic display	0632 3220
Option CO sensor without H ₂ -compensation	
Option H ₂ -compensated CO sensor	
Option CO _{low} sensor	
Bluetooth option	

Spare gas sensors	Part no.
Spare sensor O ₂ for testo 320	0393 0005
Spare CO sensor (without H ₂ -compensation) for testo 320	0393 0053
Spare CO sensor H ₂ -compensated for testo 320	0393 0105
Spare CO _{low} sensor for testo 320	0393 0103

Smoke tester with oil and soot sheet, for measuring soot in flue gas, excl. cone (part no. 0554 9010)	0554 0307	
Testo Bluetooth®/IRDA printer incl. 1 roll of printer paper, rechargeable battery and mains unit	0554 0620	
Testo fast printer IrDA with wireless infrared interface; 1 roll thermal paper; 4 AA batteries	0554 0549	
Spare thermal paper for printer, permanent ink	0554 0568	
PC analysis software easyheat, for presenting measurement procedures as diagrams, tables and for managing customer data	0554 3332	
USB mains unit incl. cable	0554 1105	
Spare battery	0515 0046	
Instrument case (height: 130 mm) for instrument, probes and accessories	0516 3300	
Instrument case with double base (height: 180 mm) for instrument, probes and accessories	0516 3301	
ISO calibration certificate/flue gas	0520 0003	

Probes

Compact basic flue gas probes		Part no.
Flue gas probe compact; length 180 mm; Ø 6 mm; Tmax. 500 °C; probe stop; NiCr-Ni thermocouple, 2.2 m hose and particle filter included	0600 9740	
Flue gas probe compact; length 300 mm; Ø 6 mm; Tmax. 500 °C; probe stop; NiCr-Ni thermocouple, 2.2 m hose and particle filter included	0600 9741	
Modular flue gas probes		Part no.
Flue gas probe modular, incl. cone for attachment; thermocouple NiCr-Ni; hose 2.2 m; particle filter; length 180 mm; Ø 8 mm; Tmax. 500 °C; TÜV-tested	0600 9760	
Flue gas probe; length 300 mm; Ø 8 mm; Tmax. 500 °C; TÜV approval; probe stop; NiCr-Ni thermocouple; 2.2 m hose and particle filter included	0600 9761	
Flue gas probe; length 180 mm; Ø 6 mm; Tmax. 500 °C; probe stop; NiCr-Ni thermocouple; 2.2 m hose and particle filter included	0600 9762	
Flue gas probe; length 300 mm; Ø 6 mm; Tmax. 500 °C; probe stop; NiCr-Ni thermocouple; 2.2 m hose and particle filter included	0600 9763	
Flue gas probe flexible; thermocouple NiCr-Ni; hose 2.2 m; dirt filter; length 330 mm; Ø 9 mm; Tmax. 180 °C; short-term 200 °C; ideal for measuring at inaccessible points	0600 9770	
Probe accessories		Part no.
Probe shaft; length 180 mm; 8 mm; Tmax. 500 °C	0554 9760	
Probe shaft; length 300 mm; Ø 8 mm; Tmax. 500 °C	0554 9761	
Probe shaft flexible; length 330 mm; Ø 9 mm; Tmax. 180 °C	0554 9770	
Probe shaft multi-hole; length 300 mm; Ø 8 mm; for mean CO calculation	0554 5762	
Probe shaft multi-hole; length 180 mm; Ø 8 mm; for mean CO calculation	0554 5763	
Hose extension; 2.8 m; extension cable for probe	0554 1202	
Hose connection set with adapter for separate gas pressure measurement	0554 1203	
Spare dirt filter, modular probe; 10 off	0554 3385	
Spare particle filter, compact probe; 10 off	0554 0040	

Probes

Additional probes

Part no.

Dual wall clearance probe for O ₂ supply air measurement	0632 1260	
Gas leak detection probe; 0 to 10000 ppm CH ₄ /C ₃ H ₈	0632 3330	
Ambient CO probe, for detecting CO in buildings and rooms; 0 to +500 ppm	0632 3331	
Ambient CO ₂ probe, Plug-in head, connection cable 0430 0143 or 0430 0145 required	0632 1240	
Differential temperature set; consisting of 2 Velcro probes and temperature adapter	0554 1208	
Fine pressure probe: highly accurate probe for the measurement of differential pressure and temperature, as well as Pitot tube measurement of flow velocities (see technical data)	0638 0330	
Capillary hose set for 4 Pa measurement	0554 1215	

Combustion air temperature probes

Part no.

Combustion air temperature probe, immersion depth 300 mm	0600 9791	
Combustion air temperature probe, immersion depth 190 mm	0600 9787	
Combustion air temperature probe, immersion depth 60 mm	0600 9797	

Additional temperature probes

Part no.

Mini ambient air probe; for separate ambient air temperature measurement; 0 to +80 °C	0600 3692	
Very fast reaction surface probe	0604 0194	
Connection cable	0430 0143	

Technical data

	Measuring range	Accuracy ± 1 digit	Resolution	Adjustment time t_{90}
Temperature	-40 to +1200 °C	± 0.5 °C (0 to +100.0 °C) ± 0.5 % of m.v. (remaining range)	0.1 °C (-40 to +999,9 °C) 1 °C (> +1000 °C)	
Draught measurement	-9.99 to +40 hPa	± 0.02 hPa or $\pm 5\%$ of m.v. (-0.50 to +0.60 hPa) ± 0.03 hPa (+0.61 to +3.00 hPa) $\pm 1.5\%$ of m.v. (+3.01 to +40.00 hPa)	0.01 hPa with fine draught option 0.001 hPa	
Pressure measurement	0 to +300 hPa	± 0.5 hPa (0.0 to 50.0 hPa) $\pm 1\%$ of m.v. (50.1 to 100.0 hPa) $\pm 1.5\%$ of m.v. (remaining range)	0.1 hPa with fine draught option 0.01 hPa	
O₂ measurement	0 to 21 Vol. %	± 0.2 Vol. %	0.1 Vol. %	< 20 s
CO measurement (without H₂-compensation)	0 to 4000 ppm	± 20 ppm (0 to 400 ppm) $\pm 5\%$ of m.v. (401 to 2000 ppm) $\pm 10\%$ of m.v. (2001 to 4000 ppm)	1 ppm	< 60 s
CO measurement (H₂-compensated)	0 to 8000 ppm	± 10 ppm or $\pm 10\%$ of m.v. (0 to 200 ppm) ± 20 ppm or $\pm 5\%$ of m.v. (201 to 2000 ppm) $\pm 10\%$ of m.v. (2001 to 8000 ppm)	1 ppm	< 40 s
Determination of degree of effectivity (Eta)	0 to 120%		0.1%	
Flue gas loss	0 to 99.9%		0.1%	
CO₂ determination digital calculation from O ₂	0 to CO ₂ max	± 0.2 Vol. %	0.1 Vol. %	
Option CO_{low} measurement (H₂-compensated)	0 to 500 ppm	± 2 ppm (0 to 39 ppm) $\pm 5\%$ of m.v. (40 to 500 ppm)	0.1 ppm	< 40 s
Ambient CO measurement (with CO probe)	0 to 500 ppm	± 5 ppm (0 to 100 ppm) $\pm 5\%$ of m.v. (> 100 ppm)	1 ppm	
Gas leak measurement for flammable gases (with gas leak detection probe)	0 to 10.000 ppm CH ₄ / C ₃ H ₈	Signal optical display (LED) audible signal via buzzer		< 2 s
Ambient CO₂ measurement (with ambient CO₂ probe)	0 to 1 Vol. % 0 to 10.000 ppm	± 50 ppm or $\pm 2\%$ of m.v. (0 to 5000 ppm) ± 100 ppm or $\pm 3\%$ of m.v. (5001 to 10000 ppm)		
Differential pressure, flow velocity and temperature via fine pressure probe	± 10.000 Pa 0.15 to 3 m/s max. -40 to +1,200 °C (dependent on probe)	± 0.5 Pa (0 to 9.99 Pa) plus ± 1 Digit $\pm 3\%$ of m.v. (10 to 10.000 Pa) plus ± 1 Digit ± 0.5 °C (-40 to 100 °C) ± 0.5 % of m.v. (rem. measuring range) plus probe accuracy	0.1 m/s 0.1 °C	

General technical data

Storage temperature	-20 to +50 °C
Operating temperature	-5 to +45 °C
Power supply	Battery: 3.7 V / 2,400 mAh Mains unit: 6 V / 1.2 A
Memory	500 measurement values

Display	Colour graphic display 240 x 320 pixels
Weight	573 g
Dimensions	240 x 85 x 65 mm
Warranty	Instrument/probe/gas sensors: 24 months Battery: 12 months

